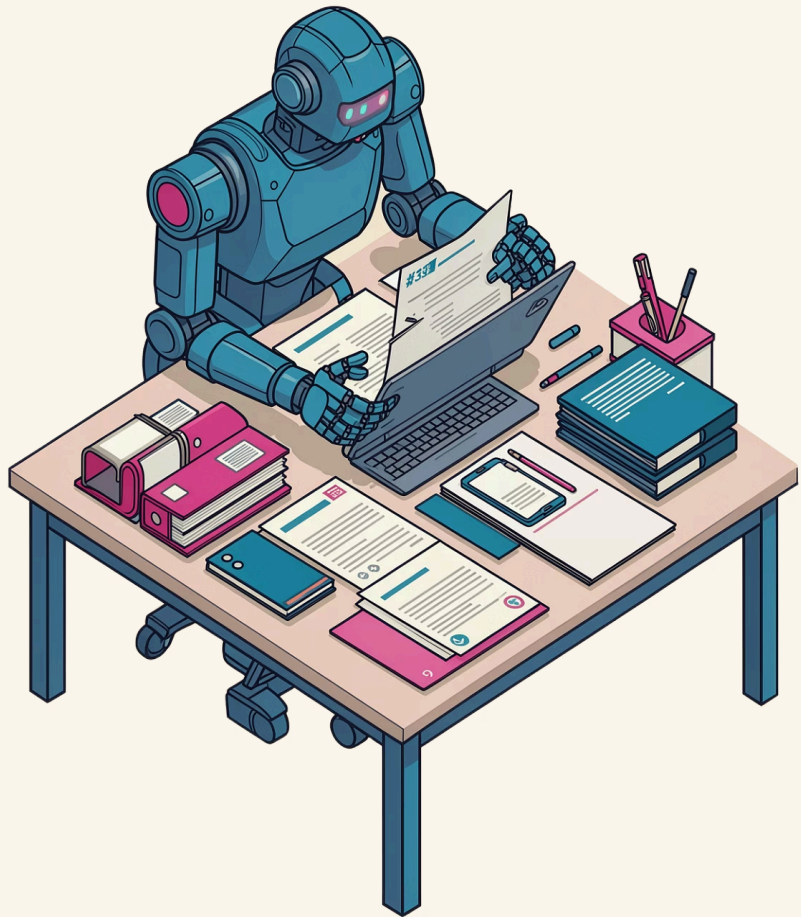




PRD Genie

An Agentic AI Assistant for Product Managers

Capstone Project 2 — Interview Kickstart | Animesh Deb

The Problem: Manual PRD Creation Wastes Time

Excessive Time Drain

PMs spend **3–5 hours** of manual drafting per transcript converting meeting notes into structured requirements.

Manual & Inconsistent

No standardization across teams — every PM follows their own process, creating quality gaps.

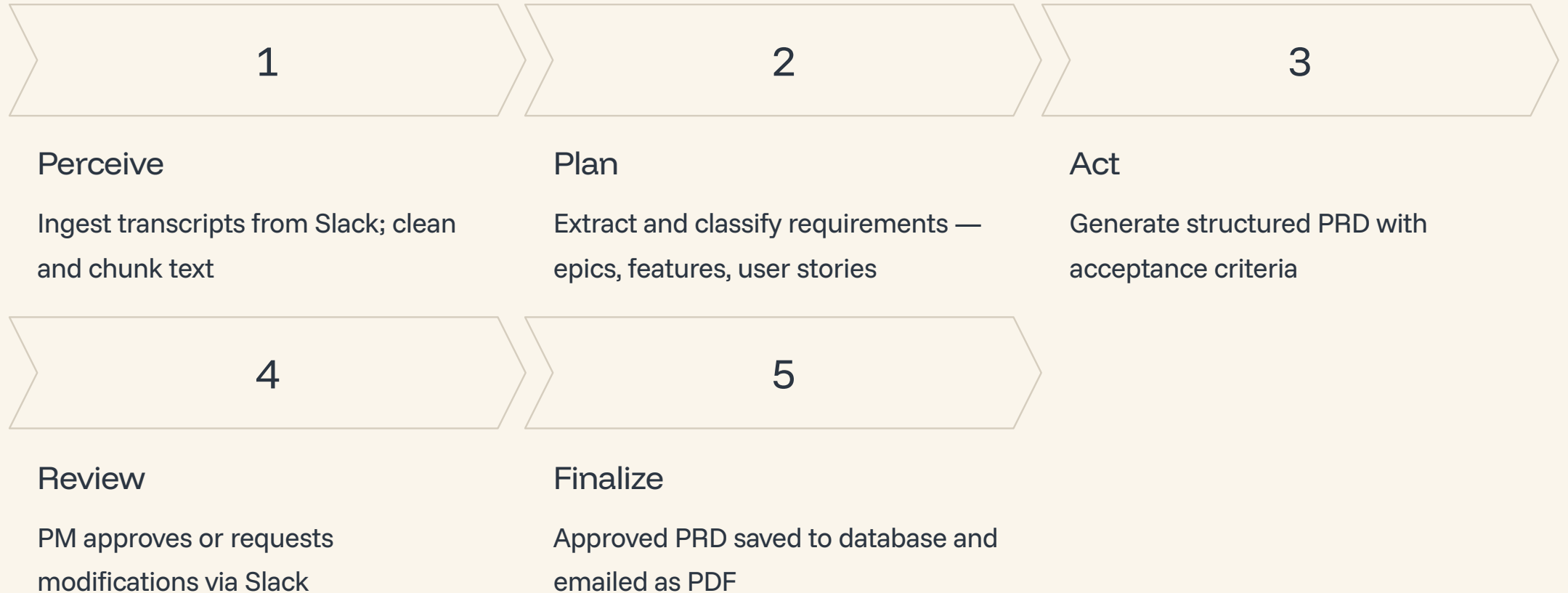
Delays Time-to-Market

Slow epic and feature breakdown stalls engineering handoff and pushes delivery timelines.



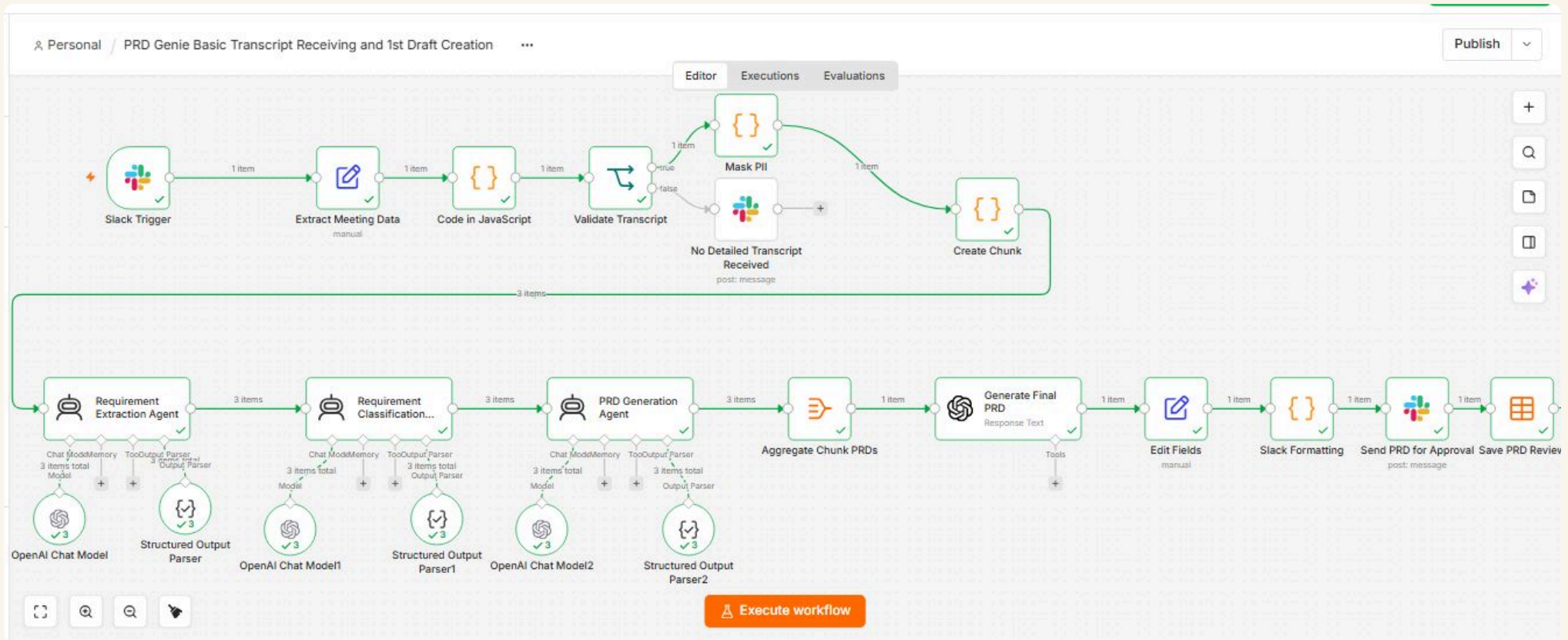
The Solution: Agentic AI Workflow

PRD Genie automates the entire requirement extraction and PRD generation pipeline using n8n, with human-in-the-loop review:



End-to-End Workflow Architecture

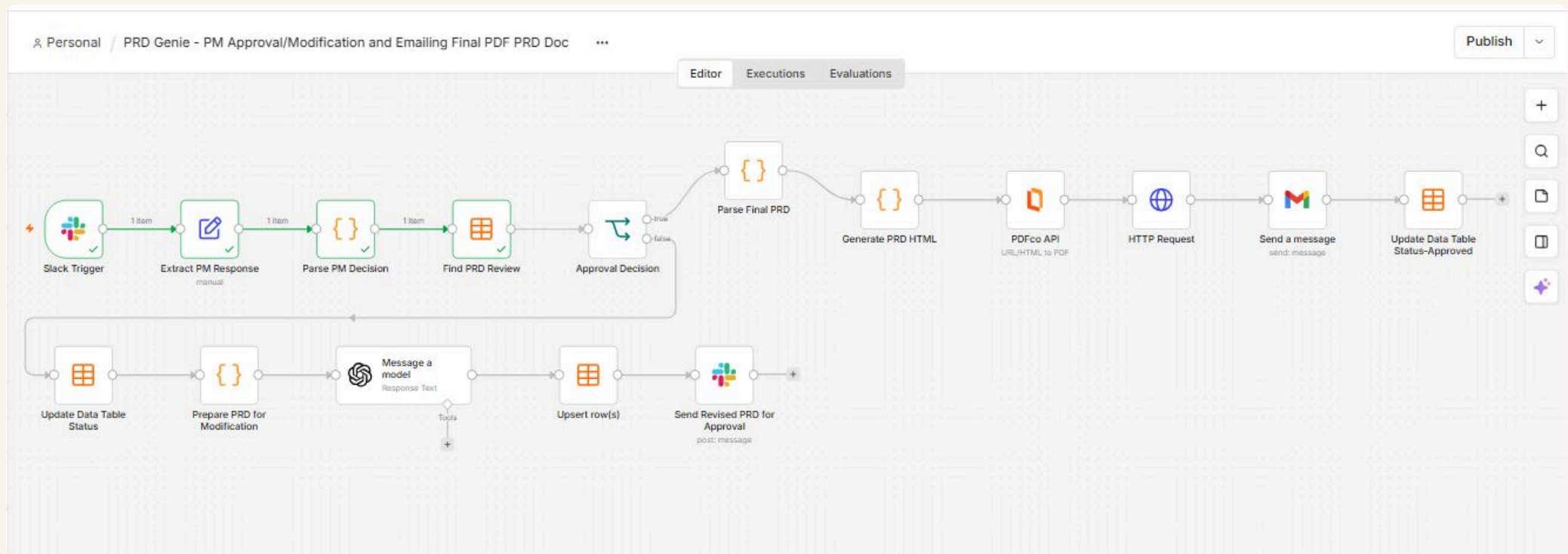
Workflow 1 - To receive transcript, extract information, draft PRD, send to PM for review and save the PRD in N8N database



N8N workflow showing data flow from Slack ingestion through extraction, classification, PRD generation, human review, and final storage in N8N database with PDF email delivery.

End-to-End Workflow Architecture

Workflow 2 - PM send approval, modifications required, PRD re-created and send to PM for approval again. Approved PRD is converted to PDF and delivered to PM via gmail. Also saved the final approved PRD to N8N database



N8N workflow showing data flow from Slack ingestion through extraction, classification, PRD generation, human review, and final storage in N8N database with PDF email delivery.

Key Workflow Stages

- Ingestion & Chunking

Receive transcripts via Slack; clean and split into manageable chunks (~2–3k tokens)

- Extraction Agent

AI extracts requirement statements, actors, goals, and constraints

- Classification

Tag each requirement as Epic, Feature, or User Story with priority

- PRD Generation

Synthesize into structured PRD with overview, epics, features, user stories, and acceptance criteria

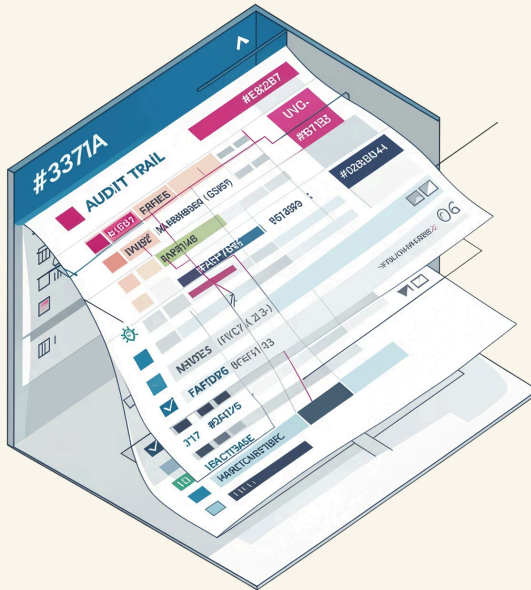
- Human Review

PM reviews draft in Slack; approves or requests modifications

- Finalization

Approved PRD stored in N8N database and emailed as PDF via Gmail

Traceability & Data Persistence



Every generated requirement is tagged with a source reference back to the originating transcript chunk, ensuring full auditability.

Chunk Table

Stores all extracted chunks with
unique IDs and source references

N8N Database

Persists PRD reviews with status tracking — **Pending** → **Modification Requested** → **Approved**

100% Traceability

Every epic, feature, user story, and constraint carries a `source_ref`, keyed by Slack thread timestamp

Chunks and Data Persistence

chunk_id	chunk_text
chunk_1	Product Manager: We want to build an AI-powered system that can detect potential major incidents before they are formally declared. \n Incident Manager: The system should monitor sudden increases in incident volume, SLA breach patterns, and configuration item instability. \n Operations Manager: False posi
chunk_2	tives are a major concern. Engineers should be able to review an alert before any action is taken. \n Product Manager: The system should provide a confidence score and explain why an incident has been flagged. \n Incident Manager: ServiceNow should be the primary source of incident information. \n Operations
chunk_3	Manager: For the MVP, the system should recommend that an engineer investigate the situation rather than automatically declare a major incident. \n Product Manager: Agreed. Human approval must remain mandatory.

Personal / Data tables / prd_reviews

2

Search

Add Row

Add Column

T prd_id	T slack_thread_ts	T meeting_title	T prd_json	T status	T pm_instruction	createdAt
msg_01385569a5e89c83006aa0236ef	1788880095.047629	AI Patient Health Monitoring & Early Ris	msg_01385569a5e89c83006aa0236ef	Modification Requested	, add observability also in the daily ope	2026-09-08T20:53:51
msg_0e04d6cd9dff398d006aa02dba6	1788882370.308259	AI Potential MI Detection Tool	msg_0e04d6cd9dff398d006aa02dba6	Pending	Null	2026-09-08T21:16:10.
msg_02ee1d6bf56e3a67006aa030974	1788883104.738509	AI Potential MI Detection Tool	msg_02ee1d6bf56e3a67006aa030974	Pending	Null	2026-09-08T21:28:24
msg_02ee1d6bf56e3a67006aa030974	1788886329.652119	AI Potential MI Detection Tool	"" *PRODUCT REQUIREMENTS DOCUM	Pending_Reapproval	, add observability also in operations	2026-09-08T22:22:12
msg_016aa20882b2ec9c006aa230db1	1789014241.506709	Smart Traffic Management AI	"" *PRODUCT REQUIREMENTS DOCUM	Pending_Reapproval	, add information of any traffic diversion	2026-09-10T09:54:01.
msg_05812e5438c86efb006aa2e9a6c	1789061552.253909	Smart Traffic Management AI	"" *PRODUCT REQUIREMENTS DOCUM	Approved	, add traffic diversion and speed limit	2026-09-10T23:02:32

Pilot Results: Exceeding Targets

70%

Cycle Time Reduction

Exceeds $\geq 50\%$ target — PM effort limited to validation, not drafting from scratch

90%

Extraction Accuracy

Average recall across sample transcripts — exceeds $\geq 85\%$ target

70%

First-Pass Approval

Drafts approved without regeneration — only wording and prioritization edits needed

100%

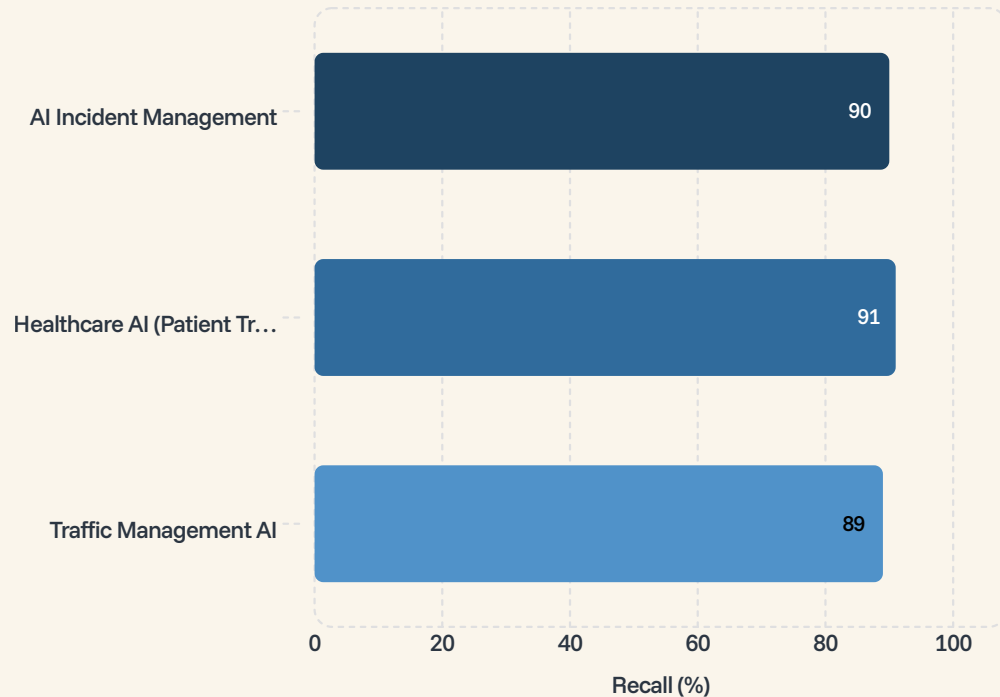
Traceability Coverage

Every generated item links back to its source transcript chunk

📄 PM edit effort averaged ~30% of draft content revised before approval — concentrated in wording and prioritization, not structure.

Sample Extraction Results

Product Domain

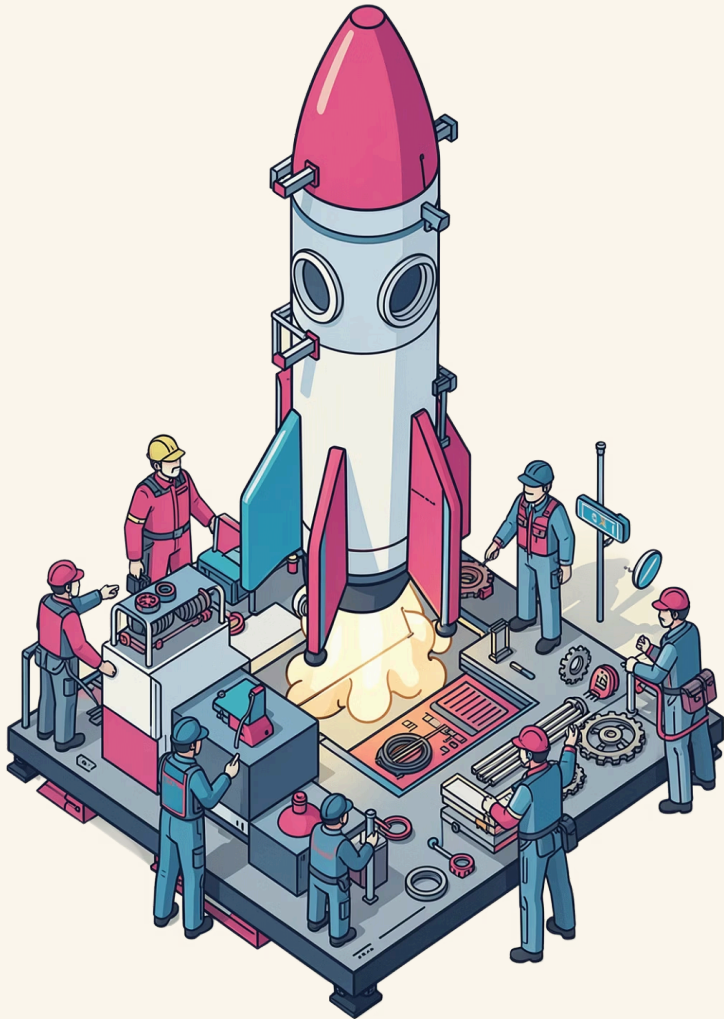


Requirement Extraction Recall Across Three Transcripts

- **AI Incident Management Platform:** 18 of 20 requirements captured — 90% recall
- **Healthcare AI Product (Patient Triage):** 20 of 22 requirements captured — 91% recall
- **Traffic Management AI Product:** 16 of 18 requirements captured — 89% recall

Average Recall: ~90% — Missed items were mostly implicit or lightly-stated requirements rather than explicitly discussed features.

Impact & Next Steps



Immediate Impact

PMs can now focus on **strategy and validation** instead of manual documentation

Scalability

Workflow is **production-ready** and handles concurrent PRD requests with minimal modifications

Continuous Improvement

Approved edits are logged to **refine prompt examples** and boost extraction accuracy over time

Team Adoption

Standardized PRD format across all teams; faster time-to-market for every new feature